

Mpemba effect

A travelling snail

September 13, 2026

1 Classical Mpemba effect

when we put two glasses of hot water and cold water into the same refrigerator, you may find the hotter one will freeze with less time. That is ridiculous. How can one state far from equilibrium state reach the equilibrium more quickly? we construct a Markov master equation:

$$\frac{dP(t)}{dt} = WP(t) \quad (1)$$

M is the transferring matrix. Obviously,

$$WP_{eq} = 0$$

we solve the eigenvalue and eigenstates of W, sorted by eigenvalues from largest to smallest.

$$0 \geq \lambda_2 \geq \lambda_3 \geq \dots$$

with

$$\lambda_1 = 0, v_1 = P_{eq}$$

Also

$$u_i^T W = \lambda_i u_i^T, u_i^T V_j = \delta_{ij} \quad (2)$$

expand the probability state with

$$P(t) - P_{eq} = \sum_{i=2} a_i(T) v_i \quad (3)$$

$$a_i(T) = u_i^T P(t) \quad (4)$$

so if somehow the magnitude of $a_2(T_{hot})$ is much smaller than $a_2(T_{cold})$, which means that the former will decay to equilibrium state faster. That's where it happens.

Energy levels and separated transitions

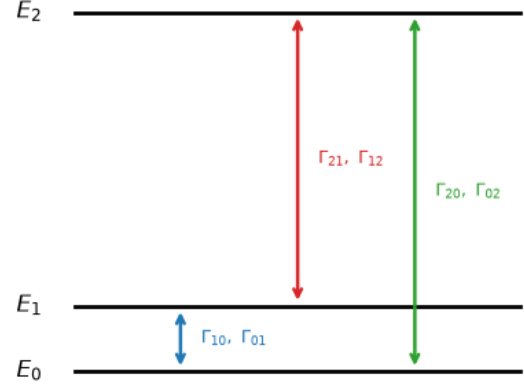


Figure 1

A simple way to verify it is to construct a three-level classical systems. Define the possibility matrix with Gibbs distribution

$$P_i(T) = \frac{e^{-\frac{E_i}{T}}}{\sum_j e^{-\frac{E_j}{T}}} \quad (5)$$

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[2] ENERGY LEVELS
E0 = 0.000000000000
E1 = 1.000000000000
E2 = 5.500000000000

[3] TEMPERATURES
Bath temperature Tb = 1.257142857143
Cold initial temperature Tc = 3.123912178045
Hot initial temperature Th = 49.931477380414
Strong-Mpemba temperature T* = 49.931477380414

[4] DOWNWARD AND UPWARD TRANSITION RATES
\Gamma_10 = 1.000000000000
\Gamma_01 = 0.451376017369
\Gamma_21 = 0.123284673944
\Gamma_12 = 0.003438208837
\Gamma_20 = 4.328761281083
\Gamma_02 = 0.054491062740
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Figure 2

$$W = \begin{pmatrix} -(\Gamma_{01} + \Gamma_{02}) & \Gamma_{10} & \Gamma_{20} \\ \Gamma_{01} & -(\Gamma_{10} + \Gamma_{12}) & \Gamma_{21} \\ \Gamma_{02} & \Gamma_{12} & -(\Gamma_{20} + \Gamma_{21}) \end{pmatrix}. \quad (6)$$

We select the numerical values satisfying detailed balancing condition.

$$\Gamma_{ij}P_i^{eq} = \Gamma_{ji}P_j^{eq} \quad (7)$$

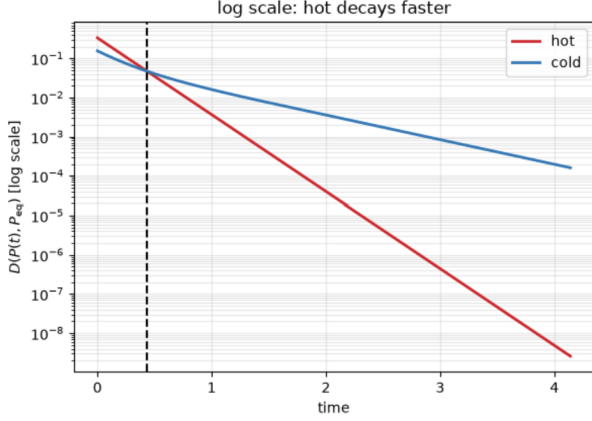


Figure 3

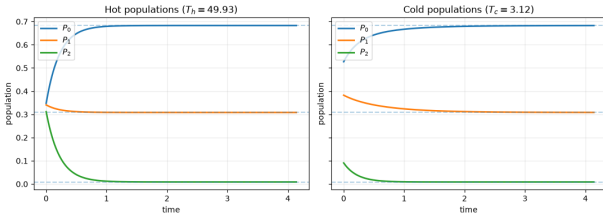


Figure 4

2 quantum Mpemba effect(example in closed system)

To study the Quantum Mpemba Effect (QME) more intuitively, we focus on the non-equilibrium dynamics of closed quantum systems. The traditional Mpemba effect typically concerns open systems coupled to an external environment, where states with different initial temperatures relax toward the same thermal equilibrium state; the anomaly lies in the fact that a state initially further from the final equilibrium can reach it sooner. The situation differs in closed quantum systems: there is no exchange of energy or information with the environment, and the system undergoes unitary evolution governed by

$$|\psi(t)\rangle = e^{-iHt}|\psi(0)\rangle$$

Consequently, the Mpemba effect cannot be simply defined by the time required for a system to dissipate into a final equilibrium state. Instead, we examine the restoration of local symmetry. We begin by preparing a series of "tilted ferromagnetic

states" with varying tilt angles, representing different degrees of broken $U(1)$ spin-rotation symmetry inherent in the Hamiltonian. Following a quantum quench, although the closed system as a whole evolves unitarily, its subsystems can undergo local relaxation and gradually restore this symmetry. In this context, the QME manifests as follows: a state with a greater initial degree of symmetry breaking achieves local symmetry restoration faster than a state with a lesser degree of breaking. We focus on XY model with attached field in z direction.

$$H = \sum_{i<j} \frac{J_{ij}}{2} (\sigma_i^x \sigma_j^x + \sigma_i^y \sigma_j^y) + \sum_i h_i \sigma_i^z \quad (8)$$

For the paper, the **"tilted ferromagnetic state"** can be written very explicitly as a tensor-product state.

Starting from the fully polarized state,

$$|\uparrow\uparrow\cdots\uparrow\rangle = |\uparrow\rangle^{\otimes N},$$

rotate every spin by the same polar angle θ and azimuthal angle ϕ . Define it as

$$|TF(\theta, \phi)\rangle = e^{-i\phi\hat{M}_z} e^{-i\theta\hat{M}_y} |\uparrow\rangle^{\otimes N}.$$

Therefore,

$$|TF(\theta, \phi)\rangle = \bigotimes_{j=1}^N \left(\cos \frac{\theta}{2} |\uparrow\rangle_j + e^{i\phi} \sin \frac{\theta}{2} |\downarrow\rangle_j \right).$$

For the case $\phi = 0$,

$$|TF(\theta, 0)\rangle = \left(\cos \frac{\theta}{2} |\uparrow\rangle + \sin \frac{\theta}{2} |\downarrow\rangle \right)^{\otimes N}.$$

for a subsystem A, We define the entanglement asymmetry

$$\Delta S_A = \ln(\text{Tr}_A(\rho_A^2)) - \ln(\text{Tr}_A(\rho_{\bar{A}}^2)) \quad (9)$$

where $\rho_{\bar{A}}$ is defined as $\sum_m \Pi_m \rho \Pi_m$ Π_m is the projection operator of eigenspace of M_z^A with eigenvalue m .EA describes how the titled state deviates from $U(1)$ symmetry .EA vanishes only if $[\rho_A, M_z^A] = 0$

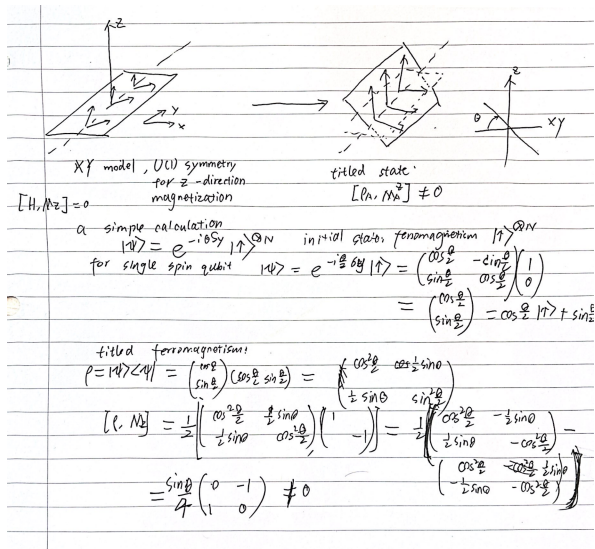


Figure 5: simple demonstration of the initial state and titled state. The direction of magnetization for the latter is rotated in Bloch sphere

2.1 macroscopic simulations

2.1.1 power law coupling

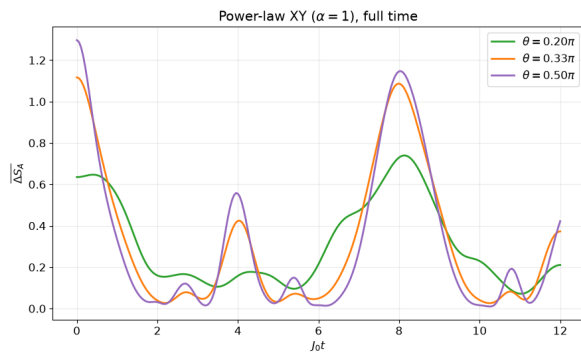


Figure 6: The dynamics of EA. Because the evolution is unitary, different lines will cross and converge for many times. We only cares about the stage $(0-3J_0t)$

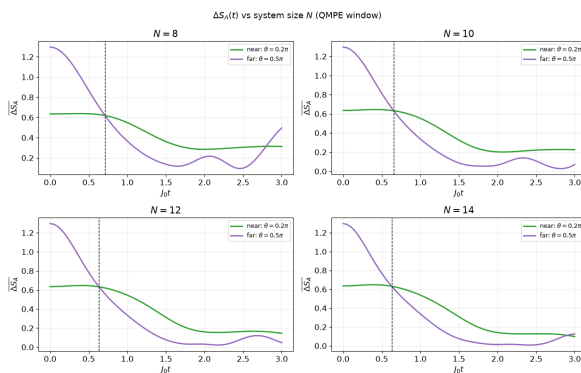


Figure 7: exclude the finite-size effect

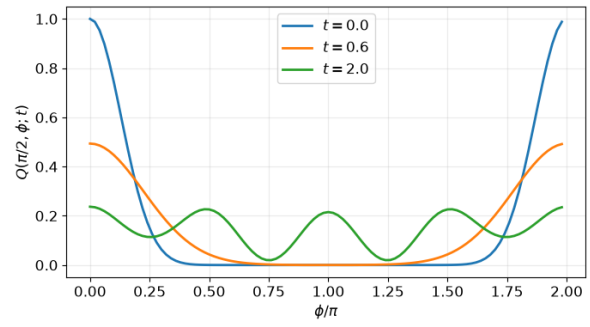


Figure 8: N=12 chain. As the symmetry getting restored, the differences in angular distribution have been smoothed out.

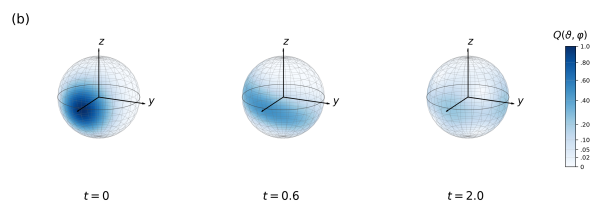


Figure 9: $Q(\phi, \theta)$ is defined as the overlap between evolved state and initial state

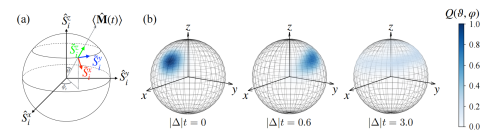


Figure 10: ideal figure

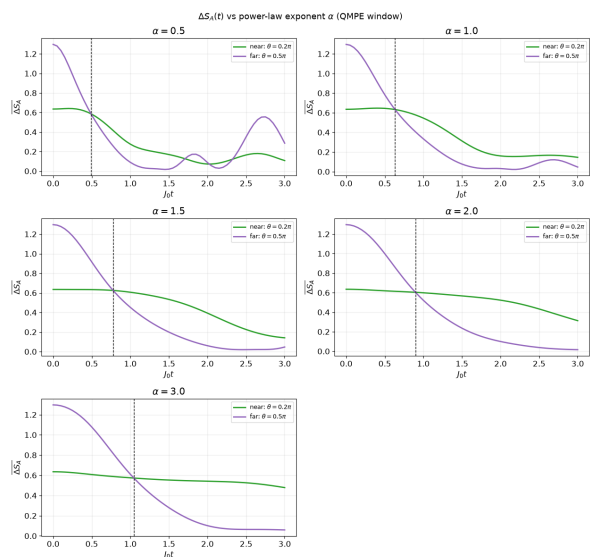


Figure 11: EA(t) with different power-law exponent α

2.1.2 exponential decaying coupling

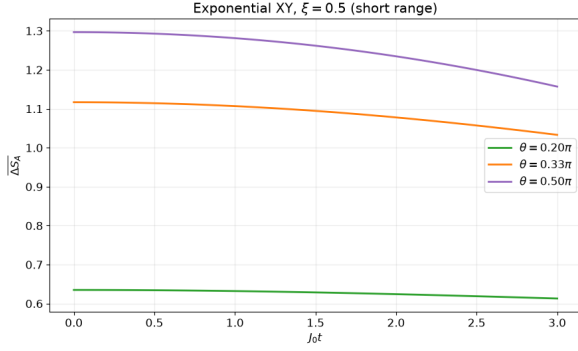


Figure 12: for $\xi=0.5$ case, we do not observe QME in finite time

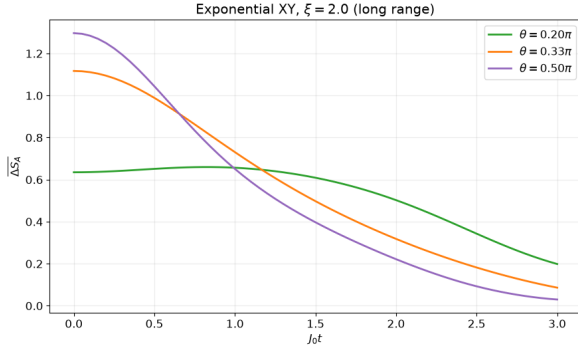


Figure 13: for $\xi = 2$ case, QME comes back unexpectedly

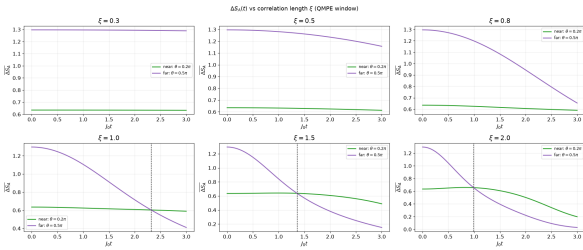


Figure 14: As the characteristic length increases, QME comes back and the crossing time is moved up

For QME in short range interaction systems, there are some different interpretations such as charge transport picture.

2.1.3 disorder

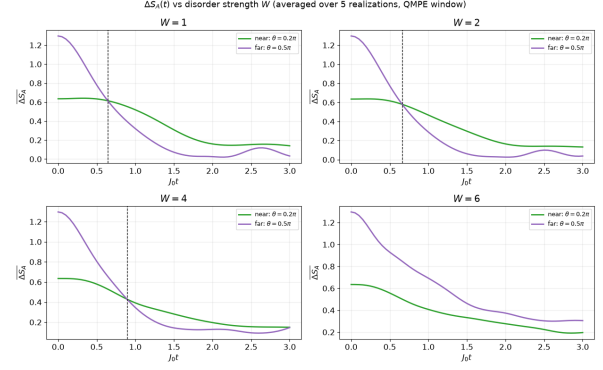


Figure 15: disorder in the z direction destroys QME

2.1.4 Gradient-ordered potential

2.2 microscopic picture: how and why ?

2.2.1 quantum fluctuations and dissipation

2.3 from closed system to open systems! Where is new physics?

for long-range model By HP transformation in the rotated frame, the initial tilted-ferromagnetic state corresponds to the bosonic vacuum state, and the quantum fluctuations of spins are described by bosonic excitations, which are referred to as spin waves.

3 Appendix

3.1 theoretical frame for long range coupling systems

To describe the fluctuations, the authors introduce a rotating frame whose $\tilde{z}$ -axis follows the instantaneous magnetization $\langle \mathbf{M}(t) \rangle$. In this frame, the large classical motion of the magnetization is separated from the quantum fluctuations around it. The Holstein-Primakoff transformation then maps these small transverse spin fluctuations onto bosonic excitations, or spin waves,

$$\tilde{S}_i^x \simeq \sqrt{\frac{s}{2}}(b_i + b_i^\dagger), \quad \tilde{S}_i^y \simeq i\sqrt{\frac{s}{2}}(b_i^\dagger - b_i), \quad \tilde{S}_i^z = s - b_i^\dagger b_i.$$

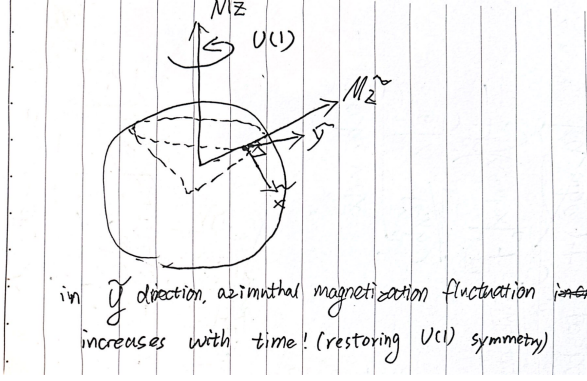


Figure 16

In this description, the initial tilted ferromagnetic state corresponds to the bosonic vacuum, while spin waves generated during the time evolution represent quantum fluctuations around the collective magnetization.

Keeping the Hamiltonian only up to quadratic order in the bosonic operators gives

$$\tilde{H} \simeq \frac{1}{2s} \sum_k \left(\xi_k b_k^\dagger b_k + \kappa_k b_k^\dagger b_{-k}^\dagger + \text{H.c.} \right),$$

with

$$\kappa_k = \tilde{J}_k \Delta \sin^2 \theta.$$

The important term is

$$\kappa_k b_k^\dagger b_{-k}^\dagger,$$

because it creates pairs of spin waves with momenta k and $-k$ from the initial bosonic vacuum. Therefore, κ_k controls the generation of quantum fluctuations. Since

$$\kappa_k \propto \sin^2 \theta,$$

a state with a larger initial tilt angle generally generates stronger spin-wave fluctuations.

The corresponding occupation of the k -th spin-wave mode is

$$n_k(t) = \frac{\Delta^2 \tilde{J}_k^2 \sin^4 \theta}{\omega_k^2} \sin^2 \left(\frac{\omega_k t}{s} \right),$$

where

$$\omega_k = \sqrt{\xi_k^2 - \kappa_k^2}.$$

Thus, the production of spin-wave fluctuations has a strong dependence on the initial tilt through the factor $\sin^4 \theta$.

For the restoration of the $U(1)$ symmetry, the most important contribution comes from the zero-momentum mode $k = 0$. The entanglement asymmetry can be approximated by

$$\Delta S_A(t) \simeq \frac{1}{2} \ln \left[\frac{2s\pi N_A \sin^2 \theta}{1 + 4n_0(t)f_A(1 - f_A)} \right], \quad f_A = \frac{N_A}{N}.$$

Therefore, as $n_0(t)$ increases, the entanglement asymmetry decreases:

$$n_0(t) \uparrow \Rightarrow \Delta S_A(t) \downarrow.$$

Since $\Delta S_A = 0$ corresponds to the restoration of the local $U(1)$ symmetry, the zero-momentum spin waves directly drive the symmetry-restoration process.

The physical meaning of n_0 becomes clearer by expressing it in terms of fluctuations of the total magnetization in the rotating frame,

$$n_0 = \frac{\text{Var}(\tilde{M}^x) + \text{Var}(\tilde{M}^y)}{2sN} - \frac{1}{2}.$$

The authors show that the fluctuation that grows during the evolution is mainly the azimuthal component,

$$\text{Var}(\tilde{M}^y).$$

Therefore, the direction of the collective magnetization becomes increasingly uncertain along the azimuthal angle ϕ . Since the original tilted ferromagnetic state breaks the $U(1)$ symmetry by selecting a particular azimuthal direction, spreading the quantum state along ϕ progressively removes this preferred direction and restores the rotational symmetry around the laboratory z -axis.

The theoretical origin of the QME can therefore be summarized as

$$\theta \uparrow \Rightarrow \kappa_0 \propto \sin^2 \theta \uparrow \Rightarrow n_0(t) \text{ grows faster} \Rightarrow \text{azimuthal quantum}$$

$$\Rightarrow \Delta S_A(t) \text{ decreases faster} \Rightarrow U(1) \text{ symmetry is restored faster.}$$

Hence, the initially more strongly tilted state, although it breaks the $U(1)$ symmetry more strongly, generates collective quantum fluctuations more rapidly and consequently restores the symmetry sooner. This provides the microscopic spin-wave explanation of the Quantum Mpemba Effect in the article.